

Recap lecture 19

NFA corresponding to Closure of FA,
Examples, Memory required to recognize
a language, Example, Distinguishing one
string from another, Example, Theorem,
Proof

Example

Let $L_{20} = \{w \in \{0,1\}^* : |w| \geq 20 \text{ and the } 20^{\text{th}} \text{ letter of } w, \text{ from right is, } 1\}$. Let S be the set of all strings of length 20, defined over Σ , any two of which are distinguishable w.r.t. L_{20} . Obviously the number of strings belonging to S , is 2^{20} . Let x and y be any two distinct strings *i.e.* they differ in i th letter, $i=1,2,3,\dots,20$, from left. For $i=1$, they differ by first letter from left.

Example continued ...

Then by definition of L_{20} , one is in L_{20} while other is not as shown below

0	.	...	.
1	.	...	.

So they are distinct w.r.t. L_{20} for $z = \square$ *i.e.* one of xz and yz belongs to L_{20} .

Example continued ...

Similarly if $i=2$ they differ by 2nd letter from left and are again distinguishable and hence for z belonging to Σ^* , $|z|=1$, either xz or yz belongs to L_{20} because in this case the 20th letter from the right of xz and yz is exactly the 2nd letter from left of x and y as shown below

.	0	...	.	z
.	1	...	.	z

Example continued ...

Hence x and y will be distinguishable w.r.t. L_{20} for $i=2$, as well. Continuing the process it can be shown that any pair of strings x and y belonging to S , will be distinguishable w.r.t. L_{20} . Since S contains 2^{20} strings, any two of which are distinguishable w.r.t. L_{20} , so using the theorem any FA accepting L_{20} must have at least 2^{20} states.

Note

It may be observed from the above example that using Martin's method, there exists an FA having $2^{20+1}-1=2,097,151$ states. This indicates the memory required to recognize L_{20} will be the memory of a computer that can accommodate 21-bits *i.e.* the computer can be in 2^{21} possible states.

Finite Automaton with output

Finite automaton discussed so far, is just associated with the RE or the language.

There is a question whether does there exist an FA which generates an output string corresponding to each input string ? The answer is yes. Such machines are called machines with output.

There are two types of machines with output. Moore machine and Mealy machine

Moore machine

A Moore machine consists of the following

1. A finite set of states $q_0, q_1, q_2, \dots$ where q_0 is the initial state.
2. An alphabet of letters $\Sigma = \{a, b, c, \dots\}$ from which the input strings are formed.
3. An alphabet $\Gamma = \{x, y, z, \dots\}$ of output characters from which output strings are generated.

Moore machine continued ...

4. A transition table that shows for each state and each input letter what state is entered the next.
5. An output table that shows what character is printed by each state as it is entered.

Moore machine continued ...

Note: It is to be noted that since in Moore machine no state is designated to be a final state, so there is no question of accepting any language by Moore machine. However in some cases the relation between an input string and the corresponding output string may be identified by the Moore machine. Moreover, the state to be initial is not important as if the machine is used several times and is restarted after some time, the machine will be started from the state where it was left off. Following are the examples

Example

Consider the following Moore machine having the states q_0, q_1, q_2, q_3 where q_0 is the start state and

$$\Sigma = \{a, b\},$$

$$\mathbb{Q} = \{0, 1\}$$

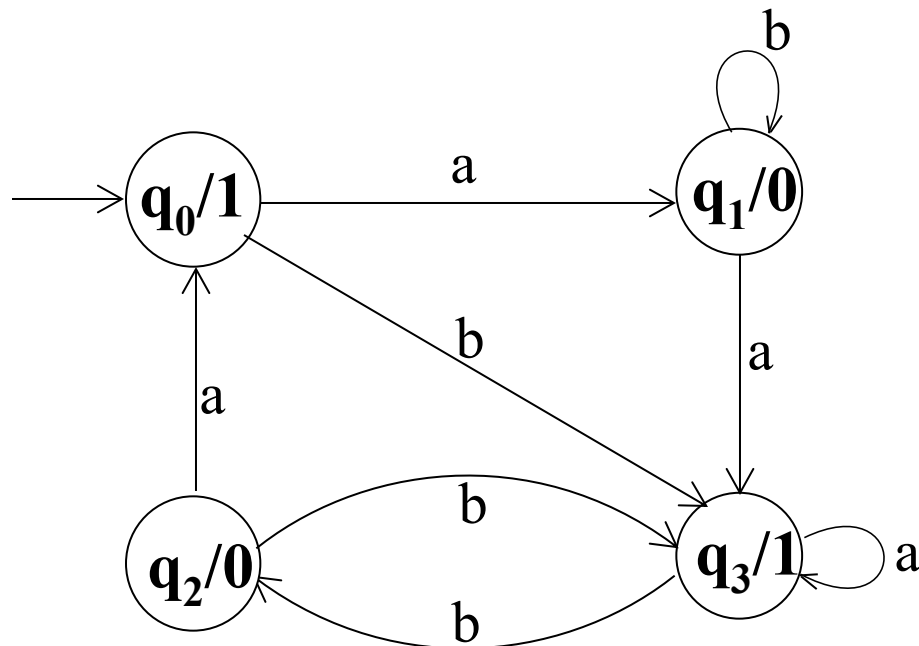
the transition table follows as

Example continued ...

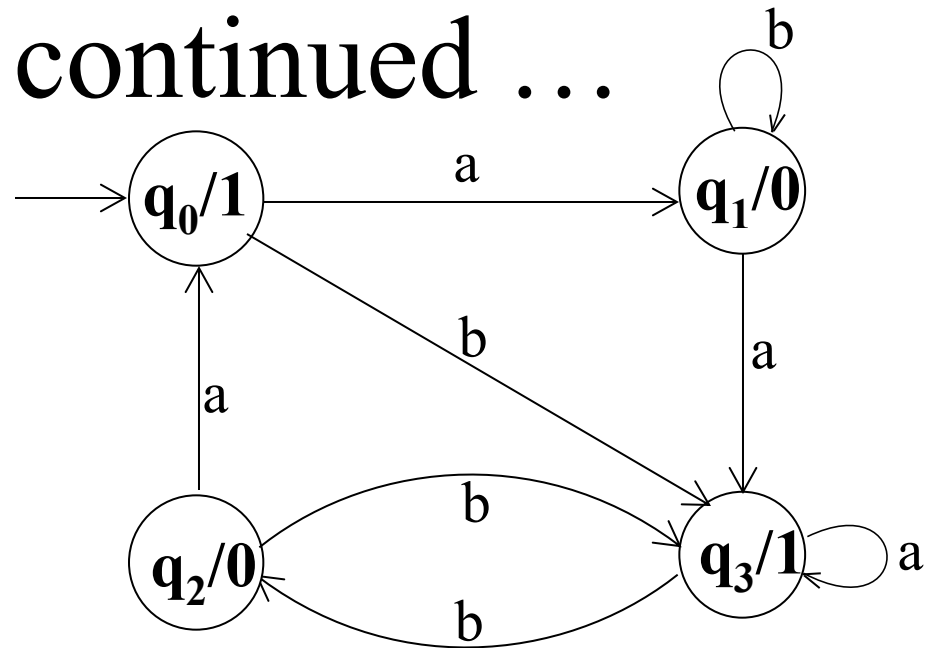
Old States	New States after reading		Characters to be printed
	a	b	
q_0^-	q_1	q_3	1
q_1	q_3	q_1	0
q_2	q_0	q_3	0
q_3	q_3	q_2	1

Example continued ...

the transition diagram corresponding to the previous transition table may be



Example continued ...



It is to be noted that the states are labeled along with the characters to be printed. Running the string abbabbba over the above machine, the corresponding output string will be 100010101, which can be determined by the following table as well

Example continued ...

Input		a	b	b	a	b	b	b	a
State	q_0	q_1	q_1	q_1	q_3	q_2	q_3	q_2	q_0
output	1	0	0	0	1	0	1	0	1

It may be noted that the length of output string is 1 more than that of input string as the initial state prints out the extra character 1, before the input string is read.

Summing Up

Recap Theorem, Example, Finite Automaton with output, Moore machine, Examples